

LINKEDLIST

Q6)

Question:

DS	Session	Linked List	Question Information	Level 1 Challenge 36
Question description A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth. As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the <code>GetNth()</code> function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position. Example Input: 1->10->30->14, index = 2 Output: 30 The node at index 2 is 30 Constraints $1 < N < 1000$ $1 < X < 1000$ $1 < I < 1000$				

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = data;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 struct Node* insertAtBeginning(struct Node *head, int data) {
17     struct Node *newNode = createNode(data);
18
19     newNode->next = head;
20     head = newNode;
21
22     return head;
23 }
24
25 int getNth(struct Node *head, int index) {
26     int count = 0;
27
28     while (head != NULL) {
29         if (count == index)
30             return head->data;
31
32         count++;
33         head = head->next;
34     }
35
36     return -1;
37 }
38
39 void display(struct Node *head) {
40     printf("Linked list:");
41
42     while (head != NULL) {
43         printf("-->%d", head->data);
44         head = head->next;
45     }
46
47     printf("\n");
48 }
49
50 int main() {
51     int N, data, index;
52     struct Node *head = NULL;
53
54     scanf("%d", &N);
55
56     for (int i = 0; i < N; i++) {
57         scanf("%d", &data);
58         head = insertAtBeginning(head, data);
59     }
60
61     scanf("%d", &index);
62
63     display(head);
64
65     printf("Node at index=%d:%d\n", index, getNth(head, index));
66
67     return 0;
68 }
```

OUTPUT:

TEST CASE 1:

```
4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:2
```

TEST CASE 2:

```
4
11 21 31 41
3
Linked list:-->41-->31-->21-->11
Node at index=3:11
```